

Unit 10, Lesson 6: Sketching Graphs of Functions

Benchmark

MA.8.F.1.3 Analyze a real-world written description or graphical representation of a functional relationship between two quantities and identify where the function is increasing, decreasing, or constant.

Learning Targets

- ☐ I can determine the independent and dependent variables being related in a functional relationship.
- ☐ I can sketch a graph from a description of a functional relationship.

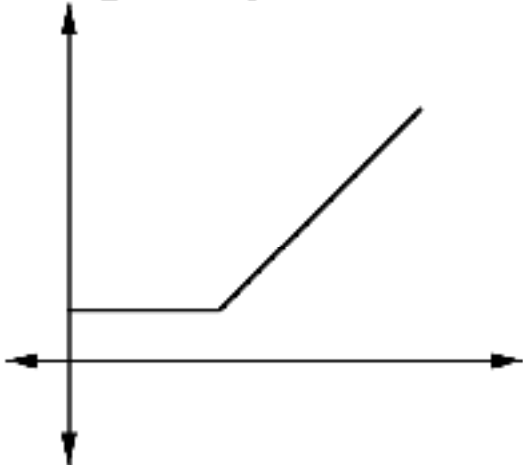
10.6.1 Warm-Up: Dog Run

Below are five pictures of a dog that were taken at equal intervals of time.

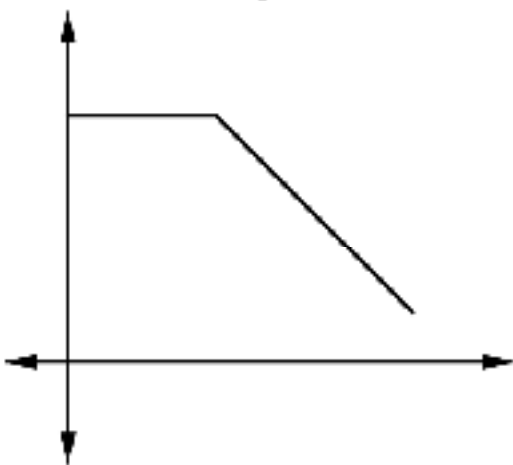


Diego and Lin drew different graphs to represent this situation.

Diego's Graph



Lin's Graph



1. They both used time as the independent variable.
What represents the dependent variable on each person's graph?

Diego's Dependent Variable	Lin's Dependent Variable

2. Explain your reasoning.

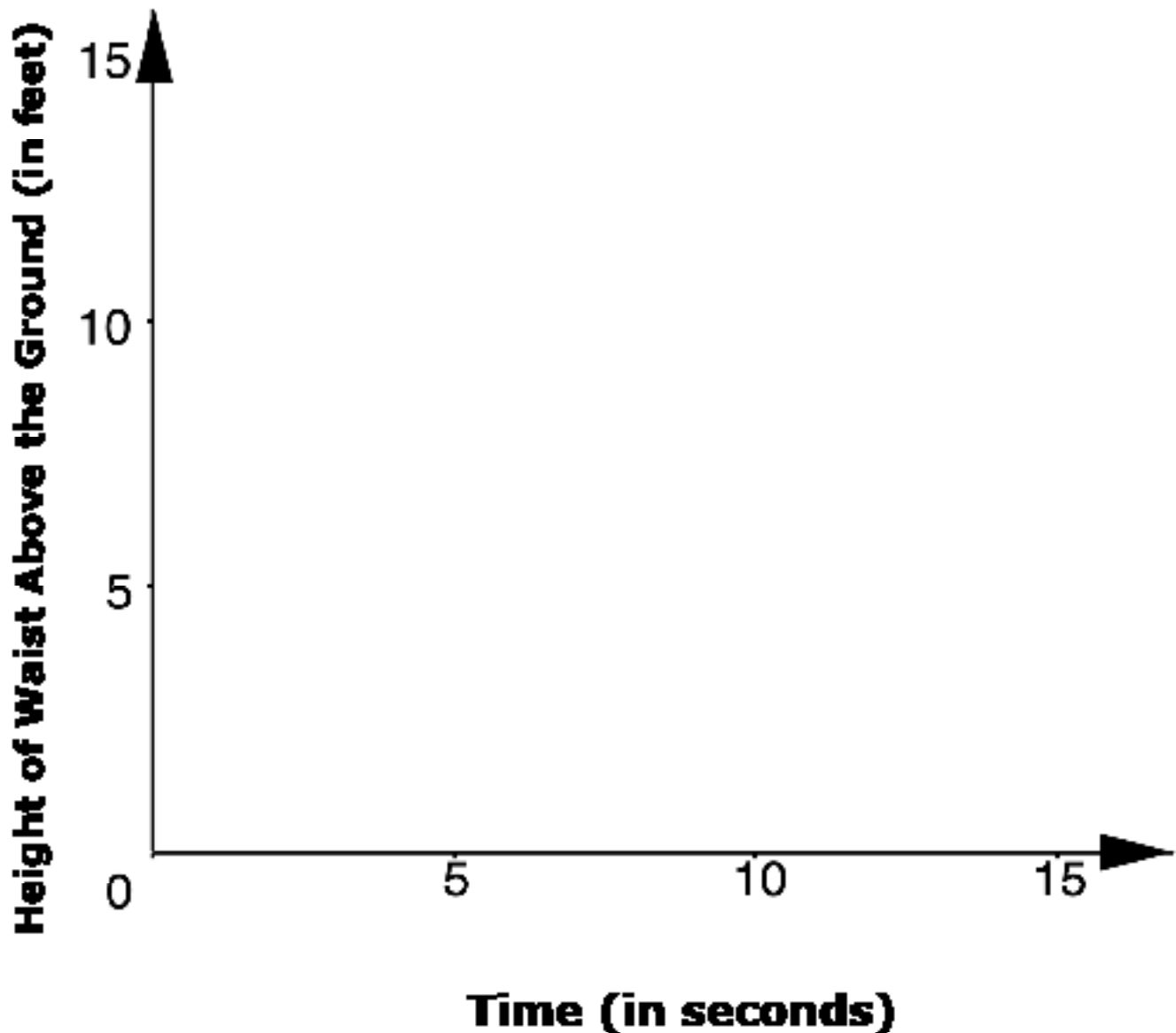
10.6.2 Activity: Graphing Stories

In this activity, several videos will be shown. If you are not able to view the videos, screenshots and brief descriptions are included for your reference.



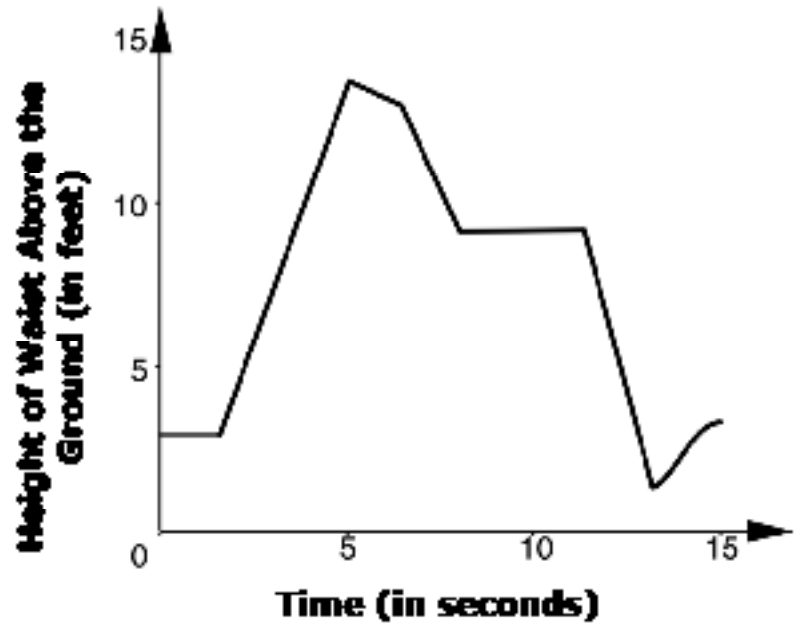
Story 1 – Playground Slide	
Screenshot	Description
	A man stands at the bottom of the ladder before climbing. He takes about 5 seconds to climb the ladder.
	The man walks across the platform at the top of the ladder to the top of the slide.
	Eight seconds after the man started to climb the ladder, he takes a seat at the top of the slide.
	He begins to slide down the slide at the 12-second mark.
	After 15 seconds have passed, the man is standing at the bottom of the slide.

1. What are some different measurements that are changing over time in this story?
2. Sketch the graph of the relationship between the height of the man's waist above the ground, in feet, and time, in seconds.



3. Compare your graph with your partner. Discuss your graphs and make any changes, if necessary.

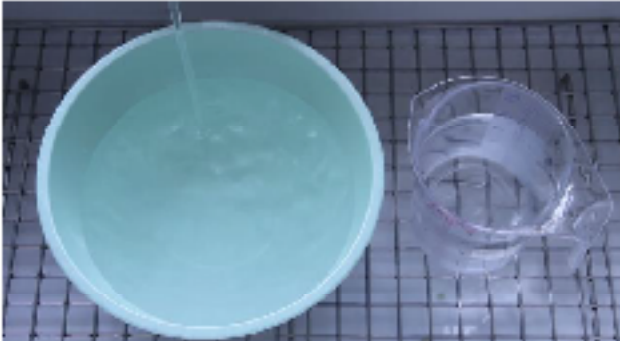
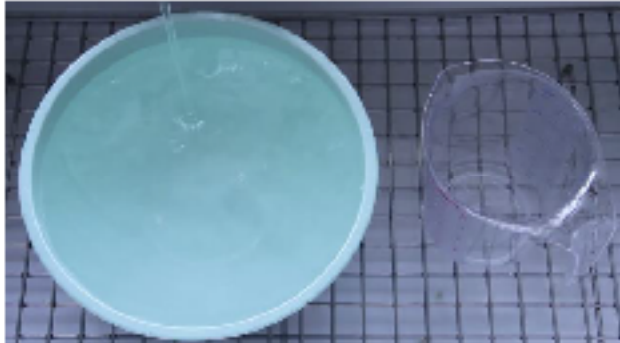
4. Peter sketched the graph shown to represent the situation.



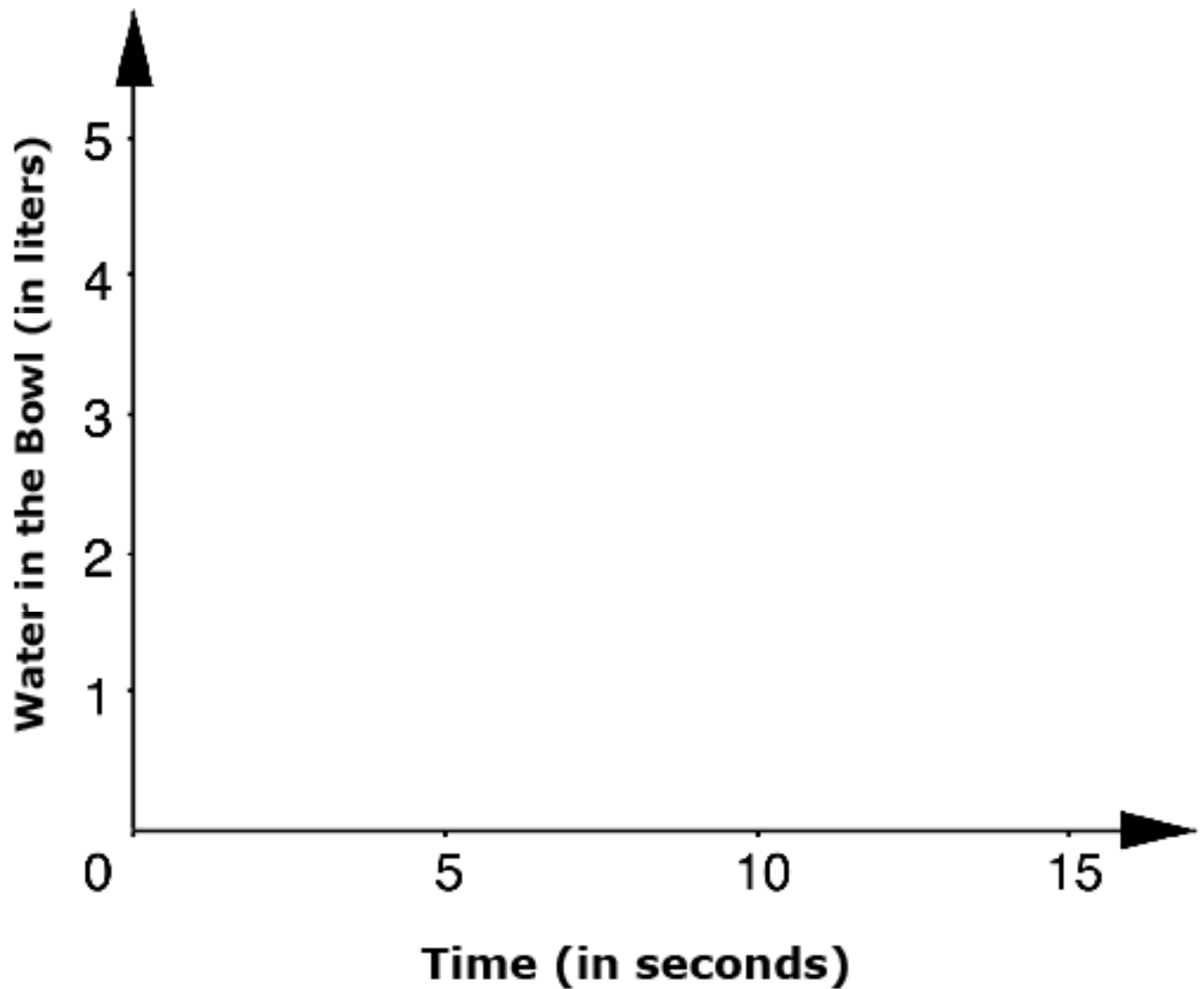
- Compare Peter's graph to your graph. Describe a feature Peter included on his graph that provides additional or different information than yours.
- What is the domain of Peter's graphical representation?
- Discuss with your partner how the sketch of the graph would change if the y-axis measured "height of the man's feet above the ground (feet)" instead of "height of waist."

Use the screenshots and descriptions for three more stories to sketch a graph representing each story.



Story 2 – Filling with Water	
Screenshot	Description
	A bowl and measuring cup each contain some water. The bowl is filled with water at a constant rate from a faucet.
	After about 5 seconds, someone pours the water from the measuring cup into the bowl as the faucet continues to run.
	The faucet continues to run for the next several seconds.

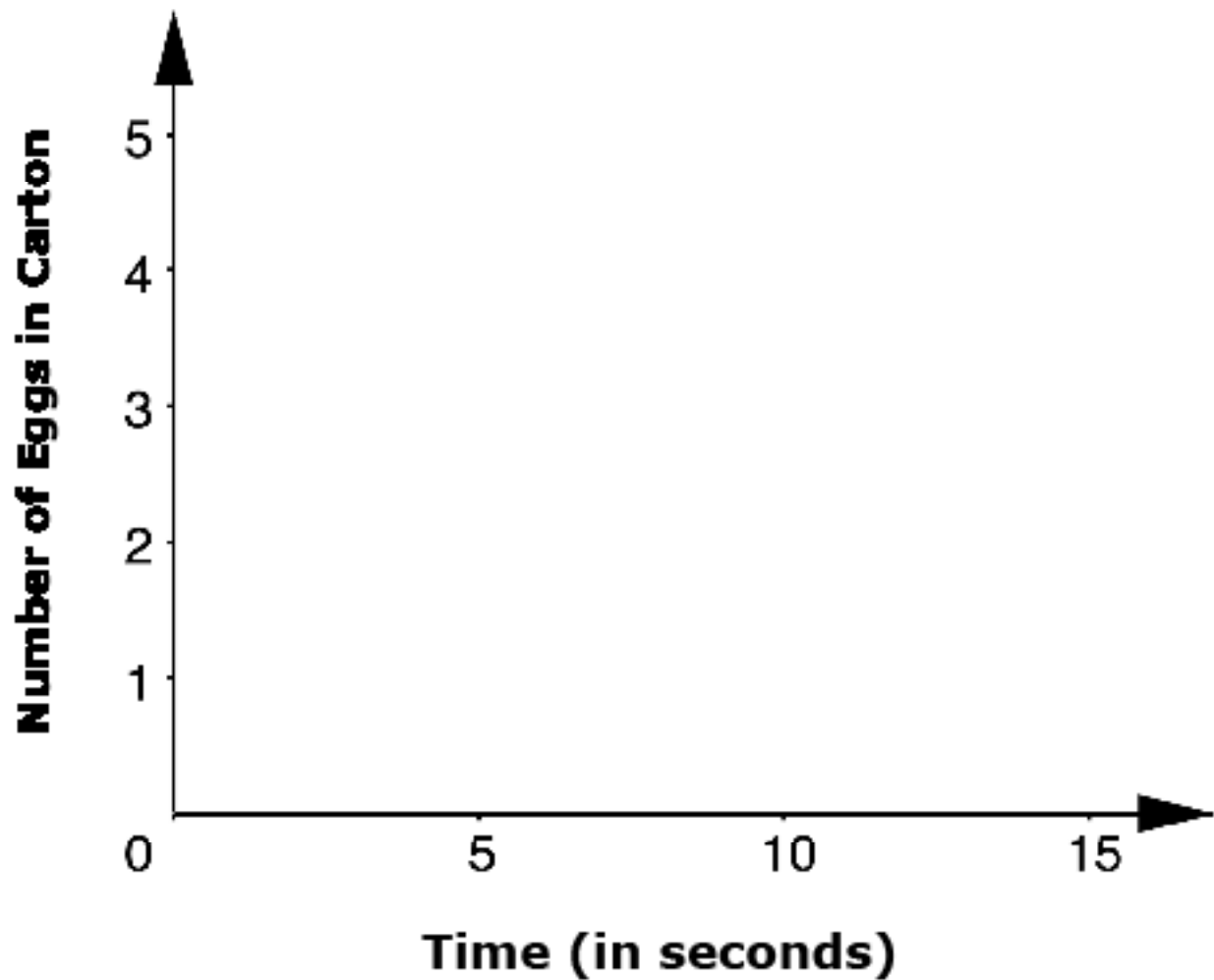
5. Sketch the graph of the relationship between the amount of water in the bowl, in liters, and time, in seconds.





Story 3 – Making Breakfast	
Screenshot	Description
	Several eggs are in an open egg carton.
	After 1 second, someone removes an egg.
	At 5 seconds, someone removes two more eggs.
	At 11 seconds, someone removes another egg.
	At 13 seconds, someone removes the last egg.

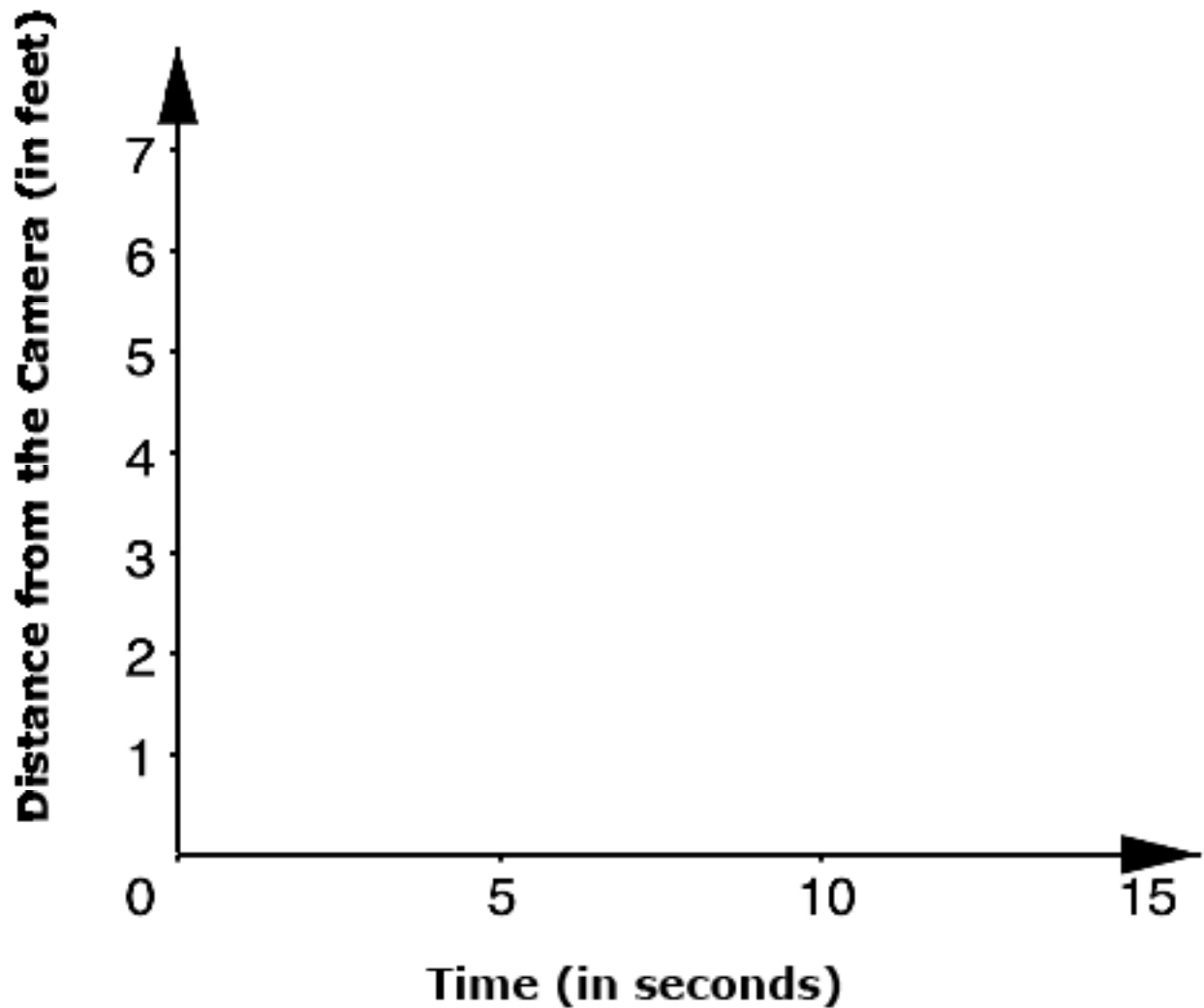
6. Sketch the graph of the relationship between the number of eggs in the carton and time, in seconds.





Story 4 – Carousel Ride	
Screenshot	Description
	A man is “riding” a carousel by pushing himself around with one foot.
	The screenshot shows the man after 1 second has passed.
	The screenshot shows the man after 2 seconds have passed.
	The man continues riding around and around on the carousel, then at 12 seconds, he gets off the carousel and walks away with his back to the camera.

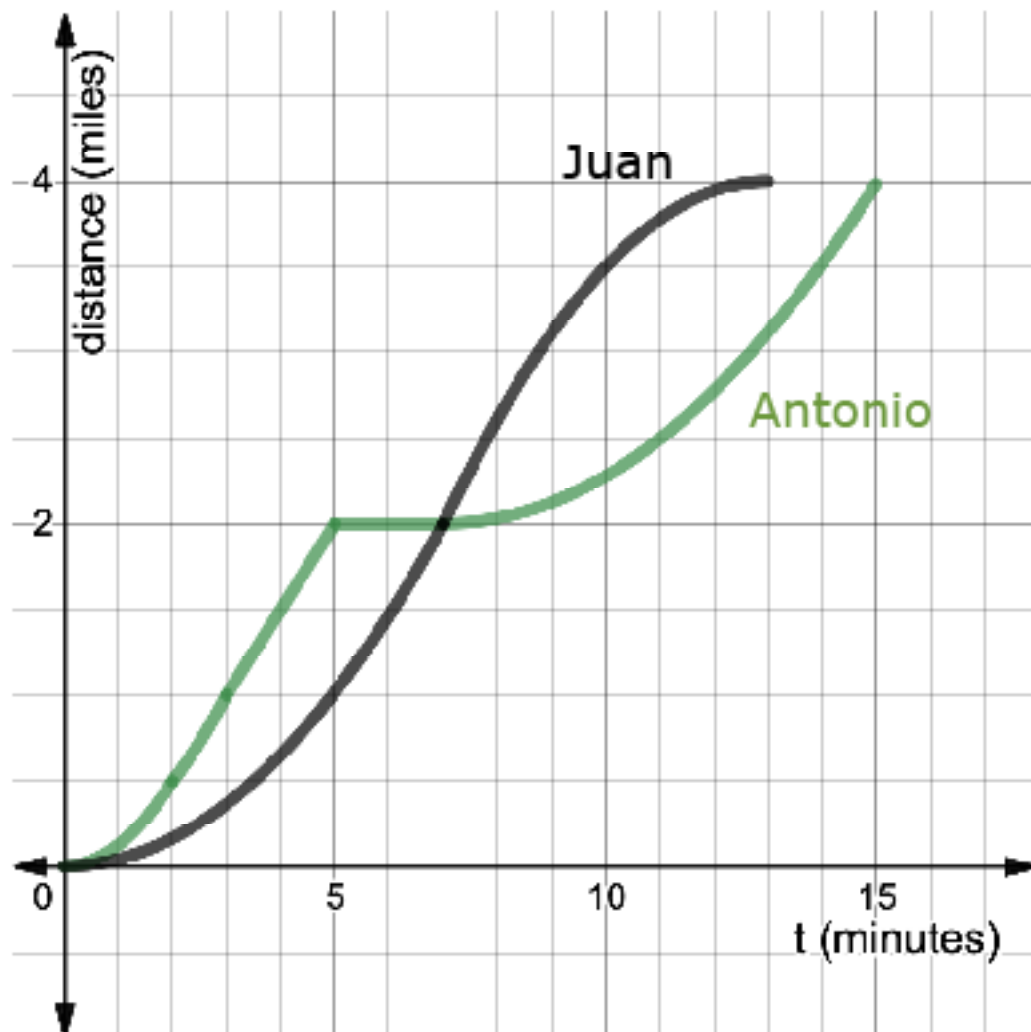
7. Sketch the graph of the relationship between the distance of the man from the camera, in feet, and time, in seconds.



8. Compare your graphs for stories 2 and 4 with your partner. Discuss any differences and make changes to your graphs based on your discussion, if necessary.

10.6.3 Try It: Create the Story

1. Antonio and Juan are in a 4-mile bike race. The graph shows the distance of each racer, in miles, as a function of time, in minutes.



- a. Who won the race? Explain your reasoning.
- b. Imagine you were watching the race and had to announce it over the radio. Write a story describing the race as a sports commentator would.

10.6.4 Wrap-Up: Cool Down

The story below describes Nima's Saturday afternoon.

"Nima was at home. He got on his bike and rode to his friend's house and stayed there for a while. Then, he rode home again. Later, he rode to the park. Then, he rode home again."

1. Sketch a graph of this story.



2. Label the axes of your graph with the names and units of measure for the two quantities being compared.
3. Based on your graph, is Nima's friend's house or the park closer to his house? Explain how you know.

